Model 1: penalized regression — methods and performance

Methods

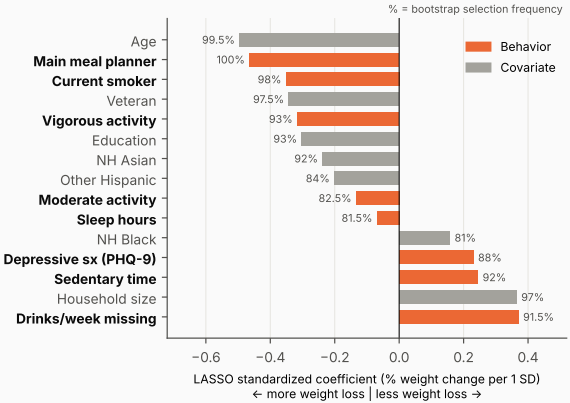
- **Models:** LASSO (L1), Ridge (L2), elastic net (L1+L2)
- **Predictor sets:** 15 covariates vs. covariates + 12 behaviors
- **Split:** 80/20 train/test (2,235/559), stratified on outcome
- **Preprocessing** (training data only): median imputation, missing flags for income, PHQ-9 and drinks/week; all predictors standardized
- **Tuning:** penalty strength by 10-fold CV
- **Added value (ΔR^2):** 2,000-sample test bootstrap + repeated 5-fold CV (25 folds)
- **Stability:** LASSO refit on 200 bootstrap samples

Performance (R^2; baseline test $R^2 = 0$)

Model	Predictors	Test R^2	CV R^2	Test RMSE
LASSO	Covariates	0.074	0.118	8.736
	+ **Behaviors**	**0.079**	**0.125**	**8.713**
Ridge	Covariates	0.072	0.116	8.750
	+ **Behaviors**	**0.079**	**0.123**	**8.715**
Elastic net	Covariates	0.074	0.117	8.736
	+ **Behaviors**	**0.079**	**0.125**	**8.713**

- Behaviors raise R^2 by $\approx$ **0.008** in CV (better in 20–21 of 25 folds)
- Test-set $\Delta R^2 = 0.005$–0.008; 95% CIs include 0 (e.g. LASSO: -0.010 to 0.021)
- All three penalties perform almost identically

Model 1: which predictors matter?



Predictors kept in $\geq$ 80% of 200 bootstrap LASSO fits.

Not shown: **BMI 1 yr ago, −3.0 per SD** (100%) — off scale.

Key findings

- **Starting BMI dominates:** $\sim 6\times$ larger than any other predictor
- **More weight loss:** main meal planner ($\approx$1 pt more), current smoking, vigorous activity
- **Less weight loss:** more sedentary time, higher PHQ-9
- Moderate activity and sleep: weaker, less consistent
- Missing drinks/week linked to less loss (likely a skip-pattern effect)
- Drinks/week, former smoking, shopping and sharing roles: rarely selected

Associations only: cross-sectional data; behaviors measured at the end of the weight-change period.